When Data Lies — Biased Dataset

Warning: Almost every "Yes" nest came from **Riverside Park** surveys. Farm and city sites were barely checked.

INSTRUCTIONS

1. Build a prediction rule from this table (as if you were training an AI).
2. Ask: would this rule work for the *whole region*?
3. Answer the discussion prompts. Then talk about how to fix the study.

Site	Region	Trees	Near river?	Playground nearby?	Nest?
1	Riverside Park	High	Yes	Yes	Yes
2	Riverside Park	High	Yes	Yes	Yes
3	Riverside Park	High	Yes	Yes	Yes
4	Riverside Park	High	Yes	No	Yes
5	Riverside Park	Medium	Yes	Yes	Yes
6	Riverside Park	High	Yes	Yes	No
7	Farm valley	Medium	No	No	?
8	Downtown	Low	No	Yes	No
9	Oak hills	High	No	No	?
10	Riverside Park	High	Yes	Yes	Yes

Our (possibly biased) rule

Discussion

1. What silly rule might an AI learn?

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2. Why would that rule fail in Oak hills?

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3. How would you redesign the study?

TEACHER REVEAL

What the “?” sites really were

Site	Region	Actual nest?	Point
7	Farm valley	Yes	Not surveyed well — model may miss real nests.
9	Oak hills	Yes	No river/playground — breaks a Riverside-only rule.

Fix ideas (after students struggle)

- Sample **many regions**, not one favorite park.
- Drop features that don’t make biological sense until tested.
- Keep humans reviewing surprising predictions.

Land this line: AI doesn’t know what’s true. It finds patterns in what we gave it.